PAGE REPLACEMENT ALGORITHM

1.FIFO

CODE :

Activities Terminal Apr 20 11:57

tejini@localhost:~/tejini/pagereplacement — /usr/bin/vim fifo.c

```c
#include<stdio.h>
int main()
{
        int i,j,n,a[50],frame[10],no,k,avail,count =0;
        printf("\nEnter number of pages: ");
        scanf("%d",&n);
        printf("Enter the page numbers (reference string):\n");
        for(i=1;i<=n;i++)
        {
                scanf("%d",&a[i]);
        }
        printf("Enter number of frames: ");
        scanf("%d",&no);
        for(i=0;i<no;i++)
        {
                frame[i]=-1;
        }
        j=0;
        printf("\nRef String\tPage Frames\n");
        printf("-----------\t----------\n");
        for(i=1;i<=n;i++)
        {
                printf("%d\t\t",a[i]);
                avail=0;
                for(k=0;k<no;k++)
                {
                        if(frame[k]==a[i])
                        {
                                avail=1;
                        }
                }
                if(avail==0){
                        frame[j] = a[i];
                        j=(j+1)%no;
                        count++;
                        for(k=0;k<no;k++){
                                if(frame[k]!=-1)
                                {
                                        printf("%d ",frame[k]);
                                }
                                else{
```

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tejini@localhost:~/tejini/pagereplacement — /usr/bin/vim fifo.c

```
        scanf("%d",&no);
        for(i=0;i<no;i++)
        {
                frame[i]=-1;
        }
        j=0;
        printf("\nRef String\tPage Frames\n");
        printf("------------\t-----------\n");
        for(i=1;i<=n;i++)
        {
                printf("%d\t\t",a[i]);
                avail=0;
                for(k=0;k<no;k++)
                {
                        if(frame[k]==a[i])
                        {
                                avail=1;
                        }
                }
                if(avail==0){
                        frame[j] = a[i];
                        j=(j+1)%no;
                        count++;
                        for(k=0;k<no;k++){
                                if(frame[k]!=-1)
                                {
                                        printf("%d ",frame[k]);
                                }
                                else{
                                        printf("- ");
                                }
                        }
                }
                else{
                        printf("  (Hit)");
                }
                printf("\n");
        }
        printf("\nTotal Page Faults: %d\n", count);
        return 0;
}
                                                                    53,1          Bot
```

OUTPUT :

Activities Terminal Apr 20 11:58

tejini@localhost:~/tejini/pagereplacement

```
[tejini@localhost pagereplacement]$ vi fifo.c
[tejini@localhost pagereplacement]$ gcc fifo.c
[tejini@localhost pagereplacement]$ ./a.out

Enter number of pages: 7
Enter the page numbers (reference string):
1 3 0 3 5 6 3
Enter number of frames: 3

Ref String      Page Frames
------------    -----------
1               1 - -
3               1 3 -
0               1 3 0
3                 (Hit)
5               5 3 0
6               5 6 0
3               5 6 3

Total Page Faults: 6
[tejini@localhost pagereplacement]$
```

2.LRU

CODE :

Activities Terminal Apr 20 12:34

tejini@localhost:~/tejini/pagereplacement — /usr/bin/vim lru.c

```c
#include<stdio.h>
int findLRU(int time[],int n){
        int i,minimum = time[0],pos=0;
        for(i=1;i<n;i++){
                if(time[i]<minimum){
                        minimum=time[i];
                        pos=i;
                }
        }
        return pos;
}
int main()
{
        int no_of_frames, no_of_pages, frames[10],pages[30],counter=0,time[10];
        int flag1,flag2,i,j,pos,faults=0;
        printf("Enter number of frames: ");
        scanf("%d",&no_of_frames);
        printf("Enter number of pages: ");
        scanf("%d",&no_of_pages);
        printf("Enter reference string: ");
        for(i=0;i<no_of_pages;i++){
                scanf("%d",&pages[i]);
        }
        for(i=0;i<no_of_frames;++i){
                frames[i]=-1;
        }
        printf("\nRef String | Page Frames\n");
        printf("------------|-------------\n");
        for(i=0;i<no_of_pages;++i){
                flag1=flag2=0;
                for(j=0;j<no_of_frames;++j){
                        if(frames[j] == pages[i]){
                                        counter++;
                                        time[j]=counter;
                                        flag1=flag2=1;
                                        break;
                        }
                }
                if(flag1==0){
                        for(j=0;j<no_of_frames;++j){
                                if(frames[j]==-1){
```

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Activities Terminal Apr 20 12:35

tejini@localhost:~/tejini/pagereplacement — /usr/bin/vim lru.c

```c
        for(i=0;i<no_of_pages;++i){
                flag1=flag2=0;
                for(j=0;j<no_of_frames;++j){
                        if(frames[j] == pages[i]){
                                        counter++;
                                        time[j]=counter;
                                        flag1=flag2=1;
                                        break;
                        }
                }
                if(flag1==0){
                        for(j=0;j<no_of_frames;++j){
                                if(frames[j]==-1){
                                        counter++;
                                        faults++;
                                        frames[j]=pages[i];
                                        time[j]=counter;
                                        flag2=1;
                                        break;
                                }
                        }
                }
                if(flag2==0){
                        pos=findLRU(time, no_of_frames);
                        counter++;
                        faults++;
                        frames[pos] = pages[i];
                        time[pos] = counter;
                }
                printf("      %d       | ",pages[i]);
                for(j=0;j<no_of_frames;++j){
                        if(frames[j]!=-1)
                                printf("%d ",frames[j]);
                        else
                                printf("- ");
                }
                printf("\n");
        }
        printf("\nTotal Page Faults = %d\n",faults);
        return 0;
}
```

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OUTPUT :

```
Activities   Terminal                    Apr 20 12:36
tejini@localhost:~/tejini/pagereplacement
[tejini@localhost pagereplacement]$ vi lru.c
[tejini@localhost pagereplacement]$ gcc lru.c
[tejini@localhost pagereplacement]$ ./a.out
Enter number of frames: 3
Enter number of pages: 7
Enter reference string: 7 0 1 2 0 3 0

Ref String | Page Frames
------------|-------------
      7       | 7 - -
      0       | 7 0 -
      1       | 7 0 1
      2       | 2 0 1
      0       | 2 0 1
      3       | 2 0 3
      0       | 2 0 3

Total Page Faults = 5
[tejini@localhost pagereplacement]$
```